Process and Design Decision

For the flooding algorithm when a node sends a new packet its assigned a source address, destination address, sequence number, TTL value, protocol type and payload. Once the packet is broadcasted to each nearby node the receiving node will check if it has seen the packet already and will drop it if it has. This makes it so there are not any loops. When the packet reaches its destination the node will process it and will stop forwarding it and will generate a ping reply if necessary. If it is not at its destination the node will decrease the TTL and then rebroadcast it. The neighbor discovery is used to discover each node's direct neighbor as. Each node will periodically broadcast a neighbor discovery packet and will record the sender in its neighbor table. The age and timer portion allows for the system to respond if a node ever leaves the network as each timer period increases the age of neighbors and will be removed if a neighbor is not heard from after some time. The age also resets if a neighbor already exists in a table. Another decision was to use a cache of the source and sequence number pairs for the flooding which allows for duplicate detection.